F3XP M.A.R.S.

A Level Physics Practice Paper 2

PHYSICS **9702/42**

Paper 4 A Level Structured Questions **August/September 2026**

MARK SCHEME

Maximum Mark: 100

Practice paper

This mark scheme goes with the practice question paper of the same number. It shows the answers and the marking points that the paper was written to test, set out in the layout of the 2025 Cambridge International mark schemes.

Alternative answers that are correct physics should be credited even where they are not listed.

Mark categories

B marks Independent marks that do not depend on other marks. The point must be seen in the answer.

M marks Method marks on which later **A** marks depend. If the **M** mark is not awarded, the **A** marks that follow cannot be awarded either.

C marks Compensatory marks. They can be awarded even if the point is not written down, when later working shows that it must have been known. If a numerical answer is correct, all the **C** marks before it are awarded automatically.

A marks Answer marks. They may depend on an **M** mark or allow a **C** mark by implication.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer.
___	Underlined content must be present in the answer to award the mark.
ecf	Error carried forward: an incorrect earlier answer used correctly later gains the later marks.
allow	Lists other answers that gain the mark.
ignore	Lists content that neither gains nor loses credit.

Question	Answer	Marks
1(a)	at minimum speed, tension in string is zero, so weight provides the centripetal force: $mg = mv^2 / r$	**M1**
	$v^2 = gr$ so $v = \sqrt{(gr)}$	**A1**
1(b)	at T: $T_T + mg = mv^2 / r$ at B: $T_B - mg = mu^2 / r$, where u is the speed at B	**C1**
	$½mu^2 = ½mv^2 + mg(2r)$ so $u^2 = v^2 + 4gr$	**C1**
	$T_B - T_T = m(u^2 - v^2) / r + 2mg = 6mg$, so the student is **not** correct the difference does not depend on v	**A1**
1(c)	minimum speed = $\sqrt{(9.81 \times 0.60)} = 2.4\ \text{m s}^{-1}$	**B1**
	greatest tension is at B: $T_T = 12.0 - 6 \times 0.15 \times 9.81 = 3.17$ N **or** $u^2 = (12.0 - 0.15 \times 9.81) \times 0.60 / 0.15 = 42.1$	**C1**
	$v^2 = (3.17 + 0.15 \times 9.81) \times 0.60 / 0.15$ **or** $v^2 = 42.1 - 4 \times 9.81 \times 0.60$ maximum speed = $4.3\ \text{m s}^{-1}$ 12.0 N applied at T, giving $7.3\ \text{m s}^{-1}$, scores B1 only	**A1**
1(d)	track can only push on the block (towards the centre), so the block must still have a speed of at least $\sqrt{(gr)}$ at the top / cannot be at rest at the top	**B1**
	$mgh = mg(2r) + ½m(gr)$ so $h = 2.5r$	**B1**
2(a)	(gravitational) potential is zero at infinity	**B1**

Question	Answer	Marks
2(a)	force is attractive, so work is done **by** the field / energy is released when a mass moves from infinity to the point **or** work must be done **on** a mass to move it from the point to infinity	**B1**
2(b)	tangent drawn to the line at $r = 8.0 \times 10^6$ m and field strength = (–) gradient	**C1**
	field strength = 0.94 N kg^{-1} allow 0.85–1.05 N kg^{-1} allow $g = -\phi / r = 7.5 \times 10^6$ / 8.0×10^6 for both marks	**A1**
2(c)(i)	$GM = -\phi r$ using a point on the line, e.g. $1.5 \times 10^7 \times 4.0 \times 10^6 = 6.0 \times 10^{13}$ N m^2 kg^{-1}	**C1**
	$GMm / r^2 = mr\omega^2$ with $\omega = 2\pi / (12.0 \times 3600) = 1.45 \times 10^{-4}$ rad s^{-1}	**C1**
	$r^3 = 6.0 \times 10^{13} / (1.45 \times 10^{-4})^2$ leading to $r = 1.42 \times 10^7$ m	**A1**
2(c)(ii)	increase in gravitational potential energy = $250 \times [(-4.2 \times 10^6) - (-1.5 \times 10^7)] = 2.7 \times 10^9$ J	**C1**
	kinetic energy in orbit = $GMm / 2r = (6.0 \times 10^{13} \times 250) / (2 \times 1.42 \times 10^7) = 5.3 \times 10^8$ J	**C1**
	energy = 3.2×10^9 J 2.7×10^9 J (kinetic energy in orbit omitted) scores C1 only	**A1**
2(c)(iii)	before launch the satellite already moves (at about 580 m s^{-1}) with the surface, in the direction of the orbit, so it already has some kinetic energy	**B1**
3(a)(i)	energy to warm all the ice to 0 °C = $150 \times 2.10 \times 18.0 = 5670$ J **and** energy to melt all the ice = $150 \times 334 = 50\,100$ J	**M1**
	energy released by the water cooling to 0 °C = $200 \times 4.18 \times 30.0 = 25\,100$ J	**M1**

Question	Answer	Marks
3(a)(i)	25 100 J < 55 800 J, so not all of the ice melts and temperature = 0 °C allow: equation assuming all the ice melts gives –21.0 °C (M1 M1), recognised as impossible for liquid water, so temperature = 0 °C (A1)	**A1**
3(a)(ii)	mass of ice melted = (25 080 – 5670) / 334 = 58.1 g	**C1**
	mass of water = 200 + 58.1 = 258 g	**A1**
3(b)(i)	q = 0 and work done on air = +12 J, so increase in internal energy = $q + W$ = +12 J	**B1**
	internal energy of an ideal gas is the total kinetic energy of its molecules, which is proportional to thermodynamic temperature, so the temperature increases	**B1**
3(b)(ii)	curve starting at (150 cm^3, 100 kPa) and lying below the given line	**B1**
	curve ending at (75 cm^3, 200 kPa)	**B1**
3(b)(iii)	temperature constant so increase in internal energy is zero; thermal energy is transferred **from** the air (to the surroundings) and is equal to the work done on the air	**B1**
	work done = area under the p–V line, which is smaller for the slow compression, so the thermal energy transferred is less than 12 J (the value is about 10 J)	**B1**
4(a)	$pV = NkT$ so $(1/3)Nm\langle c^2\rangle = NkT$	**M1**
	$\frac{1}{2}m\langle c^2\rangle = (3/2)kT$	**A1**
4(b)(i)	mean kinetic energy = $(3/2)kT$ depends only on (thermodynamic) temperature, which is the same for both, so the mean kinetic energies are equal	**B1**

Question	Answer	Marks
4(b)(ii)	ratio = $\sqrt{(32.0 / 28.0)} = 1.07$	**A1**
4(c)(i)	envelope is open, so air leaves / amount of air inside decreases (and pressure stays atmospheric): p / T is constant only for a fixed amount of gas at constant volume	**B1**
4(c)(ii)	$n = pV / RT$: initially $n = (1.01 \times 10^5 \times 2800) / (8.31 \times 288) = 1.18 \times 10^5$ mol	**C1**
	finally $n = (1.01 \times 10^5 \times 2800) / (8.31 \times 373) = 9.12 \times 10^4$ mol	**C1**
	mass = $(1.18 \times 10^5 - 9.12 \times 10^4) \times 0.029 = 780$ kg temperatures in °C score 0	**A1**
4(c)(iii)	upthrust is unchanged (weight of displaced cold air) but weight of air inside has decreased by the weight of 780 kg, so maximum mass = 780 kg ecf (c)(ii)	**B1**
5(a)	P has (centripetal) acceleration $r\omega^2$ directed towards O	**B1**
	acceleration of S is the component of this parallel to the screen: $r\omega^2 \cos\theta = \omega^2 x$, since $x = r \cos\theta$	**B1**
	$a = -\omega^2 x$ with ω constant: acceleration of S is proportional to its displacement and directed towards the equilibrium position / opposite to the displacement	**B1**
5(b)	4 maximum acceleration = $r\omega^2$	**B1**
5(c)	line below the given line at all frequencies, with a flatter (less sharp) peak	**B1**
	peak at the same or a slightly lower frequency, **not** a higher frequency	**B1**

Question	Answer	Marks
5(d)(i)	displacement of Y is maximum a quarter of a period (0.20 s) after its velocity is maximum, i.e. at t = 0.50 s; displacement of X is maximum at t = 0.20 s	**C1**
	phase difference = (0.30 / 0.80) × 360° = 135° allow 2.4 rad or 225° 45° (velocity of Y treated as its displacement) scores 0	**A1**
5(d)(ii)	$v_0 = \omega x_0$ with $\omega = 2\pi / 0.80 = 7.85$ rad s^{-1}	**C1**
	x_0 = 12 / 7.85 = 1.5 cm	**A1**
6(a)(i)	fields of A and B are in opposite directions only outside AB, and the point must be nearer the smaller charge, so it is beyond B	**B1**
	8.0 / x^2 = 2.0 / $(x – 0.30)^2$	**C1**
	x = 0.60 m 0.20 m (a point between A and B) scores 0	**A1**
6(a)(ii)	$V = Q / 4\pi\varepsilon_0 r$ for each charge, with the potentials added as scalars	**C1**
	$V = 8.99 \times 10^9 \times [(8.0 \times 10^{-9} / 0.60) – (2.0 \times 10^{-9} / 0.30)] = 60$ V ecf (a)(i)	**A1**
6(b)(i)	E = 120 / 0.025 = 4800 V m^{-1} and $a = eE / m = 8.4 \times 10^{14}$ m s^{-2}	**C1**
	$t = 0.060 / 2.30 \times 10^7 = 2.6 \times 10^{-9}$ s	**C1**
	distance = ½at^2 = 2.9×10^{-3} m	**A1**

Question	Answer	Marks
6(b)(ii)	deflection = ½(qE / m)(L / v)^2 and ½mv^2 = qV, so deflection = EL^2 / 4V, which does not depend on the mass (or charge) of the particle	**B1**
	protons are deflected by the same distance (2.9 × 10^{-3} m) but in the opposite direction (towards the 0 V plate)	**B1**
7(a)	charge per unit potential difference allow (ratio of) charge to potential difference	**B1**
7(b)(i)	charge is conserved: charge 12.0C is shared by a total capacitance 2C, so V = 12.0C / 2C = 6.0 V	**B1**
7(b)(ii)	initial energy = ½C × 12.0^2 = 72C **and** final energy = 2 × ½C × 6.0^2 = 36C	**M1**
	energy dissipated = 72C – 36C = 36C, which is half of the initial energy	**A1**
7(b)(iii)	final p.d.s, and so the final stored energy, are fixed by conservation of charge alone and do not depend on the resistance	**B1**
	so the same energy (half) is dissipated: a smaller resistance gives a larger current for a shorter time	**B1**
7(c)	gradient = –1 / τ, e.g. (–0.34 – 1.79) / 20 = –0.106 s^{-1}, so τ = 9.4 s	**C1**
	X and Y in parallel: τ = R × 2C	**C1**
	C = 9.4 / (2 × 10.0 × 10^3) = 4.7 × 10^{-4} F = 470 μF allow 450–490 μF 940 μF (total capacitance) scores C1 only	**A1**
8(a)	(magnitude of) induced e.m.f. is proportional to	**M1**
	rate of change of (magnetic) flux linkage	**A1**
8(b)	flux linkage is zero but its rate of change is maximum (at this instant)	**B1**

Question	Answer	Marks
8(b)	the sides of the coil are moving perpendicular to the field / cutting the field lines at the greatest rate, so the e.m.f. is maximum	**B1**
8(c)(i)	mean power = E_0^2 / 2R **or** $E_{r.m.s.}$ = 25 / √2 = 17.7 V	**C1**
	mean power = 25^2 / (2 × 8.0) = 39 W 78 W scores 0	**A1**
8(c)(ii)	rate of change of flux linkage doubles, so peak e.m.f. doubles to 50 V	**C1**
	mean power = 50^2 / (2 × 8.0) = 160 W allow 156 W; allow 4 × (c)(i) 39 W (only the frequency changes) scores 0	**A1**
8(d)(i)	currents in the two leads are always in opposite directions: when the current reverses, it reverses in both leads (field and current both reverse), so the force does not change direction	**B1**
8(d)(ii)	120 Hz	**B1**
9(a)	minimum wavelength (maximum photon energy) when all of the kinetic energy eV of one electron produces one photon	**B1**
9(b)	$eV = hc / \lambda_{min}$	**C1**
	$\lambda_{min} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.60 \times 10^{-19} \times 28.0 \times 10^3) = 4.44 \times 10^{-11}$ m	**A1**
9(c)	same λ_{min}, with peaks P and Q at the same wavelengths	**B1**
	greater (about double) intensity at all wavelengths	**B1**
9(d)	for peak P there must be a vacancy in the lowest level, so an electron must first be removed from the –20.0 keV level	**B1**

Question	Answer	Marks
9(d)	this needs (at least) 20.0 keV, but each electron would have only 18.0 keV	**B1**
9(e)	$\mu = \ln 2 / x_{½}$: $\mu = 0.198\ \text{cm}^{-1}$ (soft tissue) and $0.533\ \text{cm}^{-1}$ (bone) **or** path A is 2.0 half-value thicknesses; path B is 1.54 + 1.43 half-value thicknesses	**C1**
	$I_A / I_0 = 0.25$ **and** $I_B / I_0 = 0.128$	**C1**
	ratio = 0.51	**A1**
10(a)	fission of heavy nuclei and fusion of light nuclei both form nuclei with a greater binding energy per nucleon (closer to the peak near $A = 56$)	**B1**
	so in both processes the total binding energy increases / the total mass decreases and energy is released: fusion of light nuclei is not the reverse of fission of heavy nuclei	**B1**
10(b)	background subtracted: initial count rate from the isotope = 590 – 30 = $560\ \text{min}^{-1}$	**C1**
	time for the corrected count rate to halve, e.g. $560\ \text{min}^{-1}$ to $280\ \text{min}^{-1}$ (recorded $310\ \text{min}^{-1}$)	**C1**
	half-life = 6.0 h allow 5.8–6.2 h 6.5 h (background not subtracted) scores 0	**A1**
10(c)(i)	(astronomical) object of known luminosity	**B1**
10(c)(ii)	$F = L / 4\pi d^2$	**C1**
	$d = [2.0 \times 10^{36} / (4\pi \times 1.5 \times 10^{-15})]^{½} = 1.0 \times 10^{25}\ \text{m}$	**A1**
10(c)(iii)	$v = c\Delta\lambda / \lambda = 3.00 \times 10^8 \times 52 / 656 = 2.4 \times 10^7\ \text{m s}^{-1}$	**C1**

Question	Answer	Marks
10(c)(iii)	$H_0 = v / d = 2.3 \times 10^{-18}$ s^{-1} ecf (c)(ii)	**A1**